MOTION IN A STRAIGHT LINE

Introduction Point size objects are those, in which (size of object) << (Distance it moves in a reasonable duration of time.

REST If position does not change with time.
(At $t=0$, $x=0$ wrt origin O)

MOTION If position of particle changes with time.

1. TYPES OF MOTION

A) **Motion in 1-D** → If position of object changes with time w.r.t. frame of reference along a straight line, motion is 1-D.
example Motion of train along straight track.

B) Motion in 2-D → If position of object changes in plane, w.r.t. frame of reference, then motion is 2-D.

example Projectile motion

RECTILINEAR MOTION

Motion of objects along a straight line.

e.g. 1) Free fall (To drop a ball)

2) Car on a straight road

3) Train moving on a straight section of track.

4) Objects sliding on a smooth sliding surface.

5) An elevator travelling up and down between floors.

6) Motion of an object on a conveyor belt.

KINEMATICS

Study of motion of objects without knowing cause of motion.

- **Frame of reference** Motion is described relative to an origin and a set of axes. For straight line motion, we typically use an x-axis. Position to the right of origin is usually positive, and to the left is negative. (this is convention)

2. POSITION, PATH AND DISPLACEMENT

- **Position (x)** - Location of the object relative to the chosen origin.
- **Path length** - The total length of the actual path traversed by an object. It is a scalar quantity and is always positive.

- **Displacement (Δx)** - The change in position of an object. It is the straight line distance between the initial and final positions.

  - Formula $\Delta \vec{x} = \vec{x}_f - \vec{x}_i$
    (where, x_f is final position and x_i is initial position)
  - It is a vector quantity and may be positive, negative or zero.
  - Magnitude of displacement ($|\Delta \vec{x}|$) may be less than or equal to the path length.
    Equality holds only if the object moves along a straight line without changing direction.

3) VELOCITY

- **Average velocity ($\vec{v}$)**: Ratio of displacement (Δx) to the corresponding time

interval (Δt)

•• Formula $|\vec{v}| = \frac{\Delta x}{\Delta t} = \frac{x_2 - x_1}{t_2 - t_1}$

•• It is a vector quantity. It's direction is the same as the direction of displacement.

• **Average Speed** – Ratio of total path length travelled to the total time interval.

•• Formula

$$\text{Average speed} = \frac{\text{Total path length}}{\text{Total time interval}}$$

•• It is a scalar quantity.

•• Average speed is generally greater than or equal to the magnitude of average velocity.

• **INSTANTANEOUS VELOCITY**

The velocity of an object at a particular instant of time.

It is the limit of the average velocity

as the time interval (Δt) approaches zero.

•• **Calculus form**

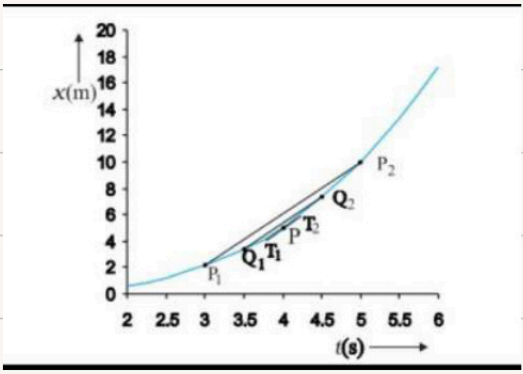
$$V = \lim_{\Delta t \to 0} \frac{\Delta x}{\Delta t} = \frac{dx}{dt}$$

•• It is the time rate of change of position.

•• **GRAPHICALLY**

It is the slope of the tangent to the position-time graph at that instant.

•• It is a vector quantity.



Tangent at any point may give velocity at that instant.

• **INSTANTANEOUS SPEED**

The magnitude of the instantaneous velocity at a given instant.

•• It is a scalar quantity.

4) ACCELERATION

- Average acceleration ($\vec{a}$)

Ratio of the change in velocity ($\Delta\vec{v}$) to the time interval (Δt) over which the change occurred.

•• Formula $\vec{a} = \frac{\Delta\vec{v}}{\Delta t} = \frac{\vec{v}_2 - \vec{v}_1}{t_2 - t_1}$

•• It is a vector quantity.

•• GRAPHICALLY

It is the slope of the straight line connecting the points $(\vec{v}_1, t_1)$ and $(\vec{v}_2, t_2)$ on a velocity-time graph.

(All graphs are shown in table)

•• INSTANTANEOUS ACCELERATION

The acceleration of an object at a particular instant of time. It is

the limit of the average acceleration as the time interval (Δt) approaches zero.

•• CALCULUS FORM:

$$a = \lim_{\Delta t \to 0} \frac{\Delta V}{\Delta t} = \frac{dV}{dt} = \frac{d^2x}{dt^2}$$

•• It is the time rate of change of velocity.

•• It is a vector quantity.

•• POSITIVE / NEGATIVE ACCELERATION

- If the acceleration has the same sign as velocity, the object is speeding up. (Positive acceleration)
- If acceleration has the opposite sign to that of velocity, the object is slowing down. (Negative acceleration or deceleration or retardation)
- The sign (+ or −) of acceleration depends on the chosen positive

direction of the axis and whether the velocity is increasing or decreasing along that direction. It does not inherently mean speeding up or slowing down without considering the velocity's sign.

- **Zero velocity or Zero acceleration**

An object may possess zero velocity momentarily (eg: at a peak of trajectory in vertical motion) while still being subjected to non-zero acceleration. This occurs because acceleration is defined as the rate of change of velocity, not the value of velocity itself.

5) MOTION WITH CONSTANT ACCELERATION

- When acceleration 'a' is constant, we can use a set of equations called KINEMATIC

EQUATIONS. (Equations may be derived graphically, also by calculus)

1) $V = V_0 + at$ (velocity as a function of time)

2) $x = x_0 + V_0 t + \frac{1}{2} a t^2$

(Position as function of time)

3) $V^2 = V_0^2 + 2a(x - x_0)$

(Velocity as function of position)

Where,

x_0 = initial position (at $t = 0$)

x = final position (at time t)

V_0 = initial velocity (at $t = 0$)

V = final velocity (at time t)

a = constant acceleration

t = time interval

Imp These equations are valid only when acceleration is constant.

(Remember to use consistent signs for displacement, velocity and acceleration based on chosen coordinate system)

- **AVERAGE VELOCITY**

(constant acceleration only)

$$\vec{v} = \frac{(\vec{v} + \vec{v_0})}{2}$$

6) **GRAPHICAL INTERPRETATION SUMMARY**

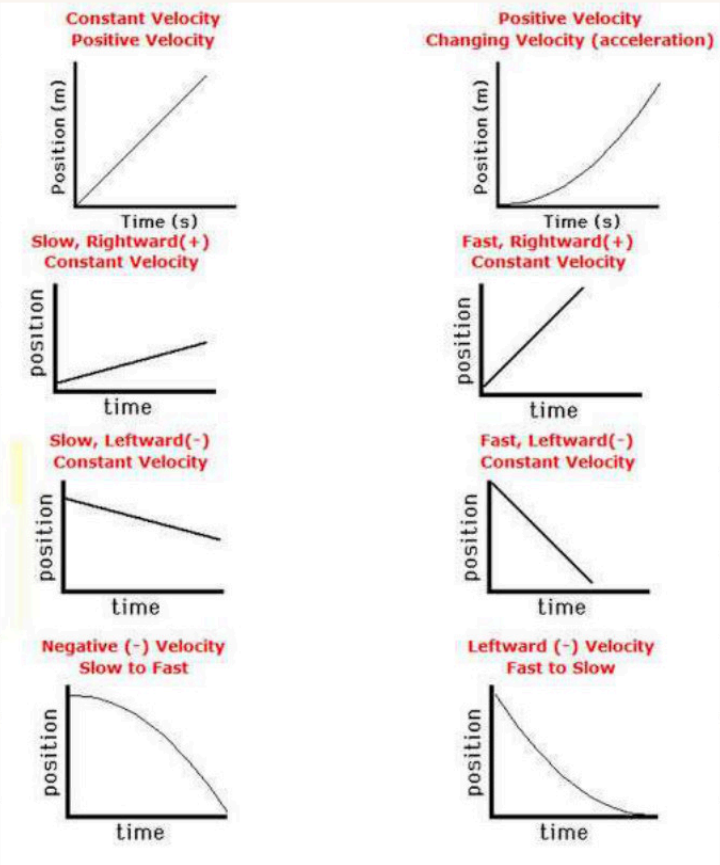
• Position - Time (x-t) Graph

.. Slope of graph = instantaneous velocity

.. Straight line parallel to time axis → Body at rest

.. straight line passing through origin or cutting an intercept on position axis → Constant velocity (zero acceleration)

.. Parabola → Constant Non-zero acceleration (curves upward for positive 'a' and downward for negative 'a'.



- VELOCITY TIME GRAPH (v-t graph)

  - Slope of tangent = instantaneous acceleration

  - Area under the curve = Displacement (Δx)

  - Straight line = constant acceleration (Zero slope = constant velocity)

7) SPECIAL CASES & APPLICATIONS

- Free fall : Motion under the influence of gravity alone (neglecting air resistance)
  - It is a case of uniformly accelerated motion near the Earth's surface.
  - Acceleration due to gravity ($g = 9.8\ m/s^2$) (directed downwards)
  - If upward direction is chosen as positive, $a = -g$

- If downward direction is chosen as negative, $a = +g$
- Kinematic equations apply with the appropriate value and sign for 'a'.

GALILEO'S LAW OF ODD NUMBERS

For a body falling from rest, the distance traversed in successive equal interval are in the ratio of odd number (1 : 3 : 5 : 7 : 9 .....).
This is the direct consequence of a kinematic equation $x = \frac{1}{2}at^2$.

- **Stopping Distance (d):** The distance, a vehicle travels after brakes are applied until it stops.

Formula

$$d = \frac{V_0^2}{2a}$$

(Where V_0 is initial velocity, a is deceleration).
Stopping distance is proportional

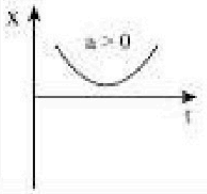
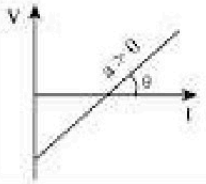
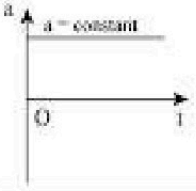
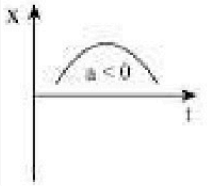
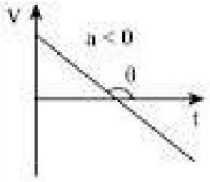
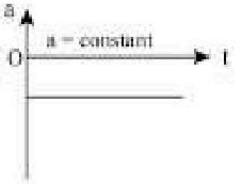
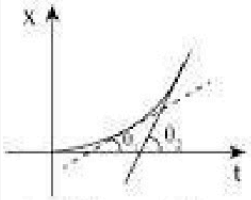
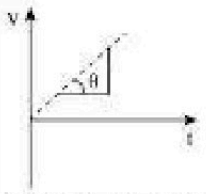
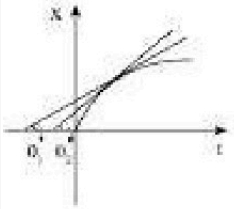
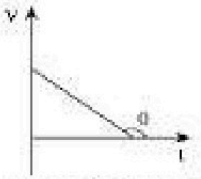
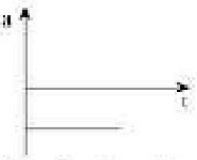
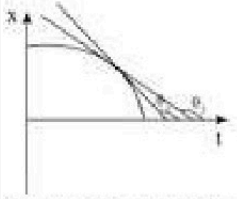
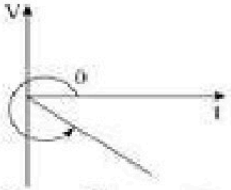
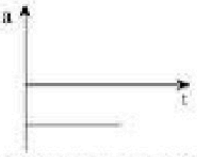
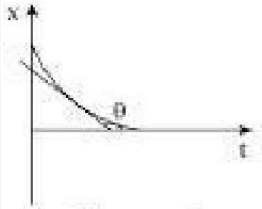
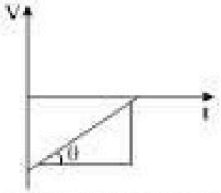
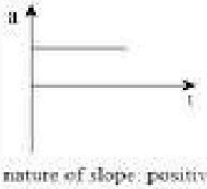
to the square of initial velocity.

•• Reaction Time (t)

The time taken by a person to observe, think and act. It can be measured using a Ruler Drop Experiment.

$$d = \frac{1}{2} g t^2$$

ALL GRAPHS FOR QUICK REVISION

S. No	Displacement – Time graph	Velocity – Time graph velocity = Slope of x – t graph	Acceleration – Time graph Acceleration = Slope of v – t graph
(i)			
(ii)			
(iii)	 nature of slope: positive magnitude of slope: Increasing	 nature of slope: positive magnitude of slope: constant	 Slope of a - t graph gives jerk, i.e., $J = \frac{da}{dt} = 0$
(iv)	 nature of slope: positive magnitude of slope: decreasing	 nature of slope: negative magnitude of slope: constant	 Slope of a - t graph gives jerk, i.e., $J = \frac{da}{dt} = 0$
(v)	 Nature of slope: negative magnitude of slope: decreasing	 Nature of slope: negative Magnitude of slope: constant	 Nature of slope: negative Magnitude of slope: constant
(vi)	 nature of slope: negative magnitude of slope: decreasing	 nature of slope: positive magnitude of slope: constant	 nature of slope: positive magnitude of slope: constant

IMP FORMULAE SUMMARY

- Average velocity

$$\vec{V} = \frac{\Delta \vec{x}}{\Delta t} = \frac{\vec{x}_2 - \vec{x}_1}{t_2 - t_1}$$

- Instantaneous velocity

$$V = \frac{dx}{dt}$$

- Average acceleration

$$\vec{a} = \frac{\Delta \vec{v}}{\Delta t} = \frac{\vec{V}_2 - \vec{V}_1}{t_2 - t_1}$$

- Intantaneous acceleration

$$a = \frac{dv}{dt}$$

- Kinematic Equations [constant acceleration only]

- $V = V_0 + at$
- $x = x_0 t + V_0 t + \frac{1}{2} a t^2$
- $V^2 = V_0^2 + 2a(x - x_0)$

* Displacement covered in n^{th} second

$$x_{nth} = V_0 + \frac{a}{2}(2n-1)$$

- Displacement

$$\Delta x = x - x_0 = V_0 t + \frac{1}{2} a t^2$$

$$= \frac{[(V + V_0)]}{2} \times t$$

- Stopping distance

$$d = \frac{V_0^2}{2|\vec{a}|}$$

Key reminder for exam

- Clearly distinguish between scalar (path length, speed) and vector (displacement, velocity, acceleration quantities)
- Pay close attention to signs (+/−) based on the chosen co-ordinate system.
- Remember that kinematic equations are applicable only for constant acceleration.
- Understand the meaning of slopes and areas in x-t and v-t graphs.
- Practice applying concepts to free fall and stopping distance problems.

All the Best

IMPORTANT POINTS FOR NEET

1) Distance $\geq$ Displacement

2) For two distances by a single object

a) for equal time interval

$$\text{average speed} = \frac{V_1 + V_2}{2}$$

if V_1 and V_2 are speeds in two different paths.

b) for equal distances

$$\text{average speed} = \frac{2V_1V_2}{V_1 + V_2}$$

3) Ratio of displacements in free fall (under gravity) in different time

$$S_1 : S_2 : S_3 : S_4 : \ldots = 1 : 4 : 9 : 16 : \ldots$$

4) for motion under gravity

Acceleration (g) is considered constant, air friction is taken to be zero and all 3 kinematic equations are valid.
for free fall, $u = 0$ (i.e. initial velocity)

Motion under gravity (from ground to ground)

- Time to go up $= \frac{u}{g}$ (u is initial speed)
- Total time of flight $= \frac{2u}{g}$
- Maximum height $= \frac{u^2}{2g}$

Calculation from graphs

- From x-t graph, slope can be calculated which gives velocity.
- From v-t graph, slope can give acceleration and area of graph with x-axis gives displacement.
- From a-t graph, slope can give jerk and area can give velocity.

Relative motion in 1-D

- Position of object 1 w.r.t. object 2

$$\vec{x}_{12} = \vec{x}_{1g} - \vec{x}_{2g}$$

x_{1g} is position of 1 w.r.t. ground

x_{2g} is position of 2 w.r.t. ground

- Similarly $\vec{V}_{12} = \vec{V}_{1g} - \vec{V}_{2g}$

- Relative motion in river (1-D)

a) Man swims upstream (opp to river)

$$\vec{V}_{mr} = \vec{V}_m - \vec{V}_r$$

where $\vec{V}_m$, $\vec{V}_r$ are velocity of man and river respectively.

b) Man swims downstream (same direction of river)

$$\vec{V}_{mr} = \vec{V}_m + \vec{V}_r$$

- Relative motion in motion under gravity

a) When 2 bodies A and B fall from same height, their motion are non uniform with constant acceleration

$$\vec{V}_{BA} = \vec{V}_B - \vec{V}_A$$

$$\vec{a}_{BA} = \vec{g} - \vec{g} = \vec{0}$$